

■ Bems Farms — Driver App & Enterprise Wallet API Specification

Complete Technical Reference: Driver Dispatch, Dedicated Virtual Accounts (DVA), Zone Earnings, Wallet Ledger & Double-Entry Accounting

Target Platforms: React Native / Flutter / iOS / Android / Web Admin	Driver Auth: JWT Bearer (`Authorization: Bearer <token>`)
Driver API Base URL: https://api.bemsfarms.com/api/driver	Admin Wallet Base URL: https://api.bemsfarms.com/api/admin/wallets
Payment Gateway: Monnify (DVA Inflows, Batch Payouts & Webhooks)	Accounting Standard: GAAP/IFRS Double-Entry General Journal

1. Driver Mobile Application Endpoints Catalog

Category	Method & Endpoint	Auth	Description
Onboarding	GET /onboarding/verify?token=...	Public	Verify invite token & driver pre-fill data
Onboarding	POST /onboarding/submit	Public	Submit driver KYC, NIN, vehicle & bank info
Auth	POST /auth/login	Public	Driver authentication & JWT token generation
Auth	POST /auth/forgot-password	Public	Request SMS/Email OTP for password reset
Auth	POST /auth/reset-password	Public	Confirm OTP code and set new password
Profile	GET /auth/me	Bearer	Retrieve driver profile, rating & Monnify DVA
Profile	PATCH /auth/profile	Bearer	Update phone, email, photo, or settlement bank
Duty	PATCH /availability	Bearer	Toggle duty status (Online / Offline)
Deliveries	GET /deliveries	Bearer	Fetch assigned active deliveries & GPS targets
Deliveries	GET /deliveries/:orderId	Bearer	Get detailed itemized order breakdown
Deliveries	PATCH /deliveries/:orderId/status	Bearer	Update milestone (picked_up, en_route, delivered)
Deliveries	GET /deliveries/history	Bearer	Paginated past delivery trips & metrics
Telemetry	POST /location	Bearer	Background GPS coordinate stream ping
Wallet	GET /earnings	Bearer	Driver wallet balance, zone pay & history
Wallet	POST /withdraw	Bearer	Request payout disbursement to personal bank

2. Driver Wallet, Dedicated Virtual Account (DVA) & Withdrawals

2.1 Get Driver Wallet & Earnings (`GET /api/driver/earnings`)

Returns live wallet balance, unwithdrawn earnings, pending payout amounts, Monnify Reserved Virtual Account (DVA) details, zone rates, and transaction ledger.

```
{
  "wallet": {
    "total_earned": 28500.00,
    "total_paid": 12000.00,
    "pending_payouts": 0.00,
    "available_balance": 16500.00,
    "commission_per_delivery": 1750.00,
    "dedicated_virtual_account": {
      "account_number": "8559127267",
      "bank_name": "Monnify / Wema Bank",
      "account_name": "BEMS - VICTOR KALU",
      "is_frozen": false
    },
    "settlement_bank": {
      "bank_name": "GTBank",
      "account_number": "0123456789",
      "account_name": "Victor Kalu"
    }
  },
  "zone_rates": [
    { "zone_id": "ZONE001", "zone_name": "Maitama & Asokoro", "driver_earning_fee": 700.00, "commission_percent": 70 },
    { "zone_id": "ZONE002", "zone_name": "Garki & Wuse", "driver_earning_fee": 1750.00, "commission_percent": 70 },
    { "zone_id": "ZONE006", "zone_name": "Interstate Express", "driver_earning_fee": 24500.00, "commission_percent": 70 }
  ]
}
```

```

    ],
    "recent_commissions": [
      { "id": 104, "total_earned": 1750.00, "deliveries": 1, "status": "pending", "created_at": "2026-09-21T09:10:00Z" }
    ],
    "recent_payouts": [
      { "payout_ref": "PAY-MNFY-8819", "amount": 12000.00, "status": "paid", "processed_at": "2026-09-20T14:30:00Z" }
    ]
  }
}

```

2.2 Request Payout / Withdrawal (POST /api/driver/withdraw)

Request Payload	Success Response (200 OK)
<pre> { "amount": 10000.00, "bank_name": "GTBank", "account_number": "0123456789", "account_name": "Victor Kalu", "notes": "Weekly earnings payout" } </pre>	<pre> { "message": "Withdrawal request of ₦10,000.00 submitted successfully!", "payout_ref": "PAY-MNFY-99201", "available_balance": 6500.00, "status": "pending" } </pre>

3. Admin Enterprise Wallet Hub & Monnify Gateway APIs

Category	Method & Endpoint	Auth Role	Description
Overview	GET /api/admin/wallets/summary	Superadmin / Accountant	Fleet floating liability, Monnify reserve (₦5B), queue size
Driver Fleet	GET /api/admin/wallets/drivers	Superadmin / Manager	List driver balances, unwithdrawn pay, DVA status, freeze state
Payout Queue	GET /api/admin/wallets/payouts	Superadmin / Accountant	Paginated driver withdrawal requests (pending, paid, rejected)
Disbursement	POST /api/admin/wallets/payouts/:id/disburse	Superadmin / Accountant	Instant bank transfer via Monnify API + General Ledger debit
Batch Payout	POST /api/admin/wallets/payouts/batch-disburse	Superadmin	Batch disburse multiple approved driver payouts in one call
Approval	PATCH /api/admin/wallets/payouts/:id/status	Superadmin / Manager	Approve or reject payout request with reason notes
Adjustment	POST /api/admin/wallets/adjust	Superadmin	Manual driver credit/debit adjustment (bonus or penalty) + Journal
Security	PATCH /api/admin/wallets/drivers/:id/freeze	Superadmin	Freeze or unfreeze driver wallet with security hold reason
Gateway	GET /api/admin/wallets/gateway/transactions	Superadmin / Accountant	Customer storefront Monnify checkouts (Gross, Fee 1.5%, Net)
Webhooks	GET /api/admin/wallets/gateway/webhooks	Superadmin	Monnify inbound webhook audit logs with SHA-512 signatures

3.1 Single Driver Payout Disbursement Payload (POST /api/admin/wallets/payouts/:id/disburse)

Request Payload	Success Response (200 OK)
<pre> { "disbursement_method": "monnify_transfer", "notes": "Approved September Batch #1" } </pre>	<pre> { "message": "Payout of ₦12,000.00 successfully disbursed!", "payout": { "id": 88, "payout_ref": "PAY-MNFY-99201", "status": "paid", "gateway_reference": "MNFY-DISB-M88X9A", "session_id": "9990581789211094", "bank_response_code": "00" } } </pre>

4. Double-Entry General Ledger Rules for Wallets

All wallet transactions post automatically to the central Chart of Accounts (COA), maintaining zero variance across the company balance sheet.

Transaction Event	Debit Entry (Dr)	Credit Entry (Cr)	Ledger Sub-System
Customer Online Checkout	1120 Monnify Settlement Vault	4110 Sales Revenue	Central General Journal
Order Delivery Completion	5210 Delivery Commission Expense	2120 Driver Wallet Payable	Driver Wallet Ledger + COA
Driver Bank Disbursement	2120 Driver Wallet Payable	1120 Monnify Settlement Vault	Central General Journal
Driver Bonus Award	5220 Driver Bonus Expense	2120 Driver Wallet Payable	Driver Wallet Ledger + COA
Driver Penalty Deduction	2120 Driver Wallet Payable	4210 Penalty & Other Revenue	Driver Wallet Ledger + COA

5. Zone-Based Compensation Rate Card (Zones 1–6)

Zone ID	Zone Coverage Area	Customer Delivery Fee	Driver Drop Payout (70%)	Estimated ETA
ZONE001	Zone 1 - Maitama & Asokoro	■1,000.00	■700.00	20 - 35 mins
ZONE002	Zone 2 - Garki, Wuse, Central Area	■2,500.00	■1,750.00	25 - 45 mins
ZONE003	Zone 3 - Utako, Jabi, Mabushi	■3,500.00	■2,450.00	30 - 50 mins
ZONE004	Zone 4 - Gwarinpa, Kubwa Expressway	■5,000.00	■3,500.00	40 - 65 mins
ZONE005	Zone 5 - Lugbe, Airport Road	■8,000.00	■5,600.00	45 - 75 mins
ZONE006	Zone 6 - Interstate / Regional Express	■35,000.00	■24,500.00	Same Day / Next Day